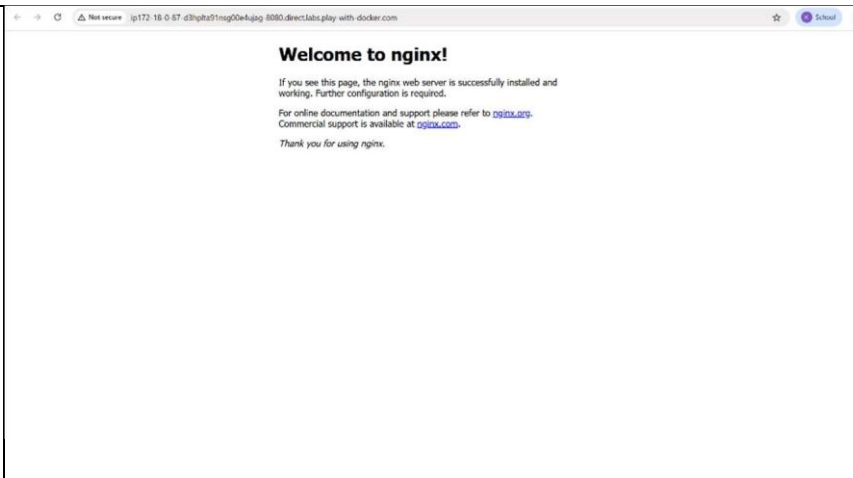


Q. No.	Question	Expected Time to complete
1	Topic: “From Hello-World to Real World – Docker for Cloud Engineers” Objective: To explore Docker’s official getting-started flow, understand container basics, and reflect on how containers simplify cross-platform development.	06-10-2025 EOD
	Task 1 – Explore Docker Hub & Getting Started 1. Visit https://www.docker.com/get-started/ and refer Get started Docker Docs 2. Read the short intro: “What is Docker & Why use it?” Docker is a platform that enables developers to build, ship, and run applications inside containers. Containers are lightweight, portable, and consistent across environments, making it easier to develop and deploy software without worrying about system differences. 3. Explore the Docker Hub (https://hub.docker.com) and note: o one official image you found useful (e.g., nginx, mysql, python) o its pull command (docker pull nginx) o the purpose of that image in industry. Image Name: nginx Pull Command: docker pull nginx Purpose in Industry: NGINX is widely used as a web server, reverse proxy, and load balancer. It's known for its high performance and scalability, making it a popular choice for hosting websites, serving APIs, and managing traffic in cloud-native applications.	



Task 2 – Hands-on Quick Test (Online Lab or Local)

Use Play-with-Docker → <https://labs.play-with-docker.com/>

Run these commands inside the browser:

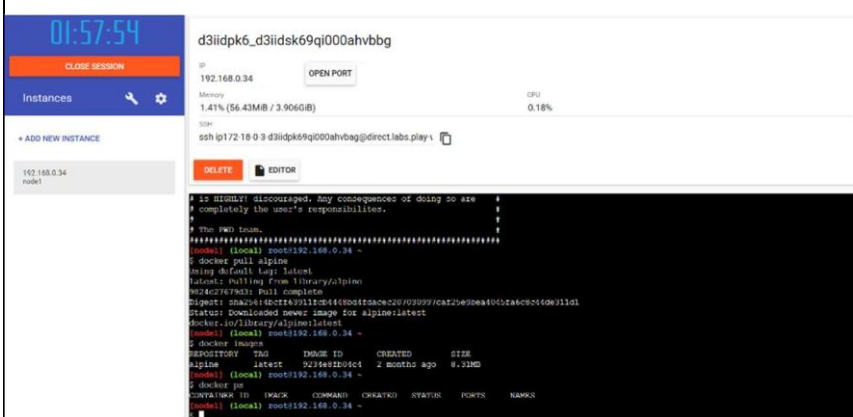
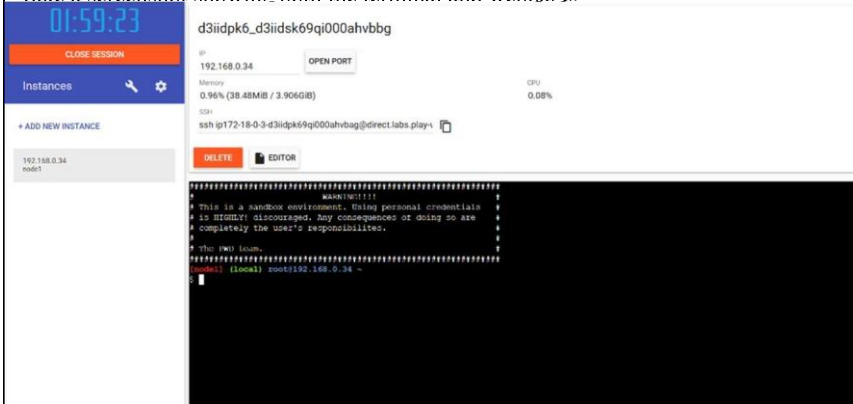
docker pull alpine

docker images

docker ps

Then open the port 8080 preview to see the NGINX welcome page.

Take a screenshot showing both the terminal and webpage.



01:56:14

CLOSE SESSION

Instances

+ ADD NEW INSTANCE

192.168.0.34
model

d3iidpk6_d3iidsk69qi000ahvbagg

IP: 192.168.0.34

OPEN PORT 8080

Memory: 7.15% (286.1MB / 3.90GB)

CPU: 0.07%

SSH: ssh ip172-18-0-3-d3iidpk69qi000ahvbagg@direct.labs.play-i

DELETE EDITOR

```

$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
alpine        latest    92f4e01b0c4    2 months ago  8.11MB
$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS   NAMES
$ docker run -d -p 8080:80 nginx
Unable to find image 'nginx:latest' locally
alpine: Pulling from library/nginx
5f25515e26d0: Pull complete
4d8505b0a0d0: Pull complete
0b0cf427d0d4: Pull complete
3cc5f5d3117a: Pull complete
4f40e020f03d: Pull complete
Digest: sha256:f4addd3b69e2674678ee2c7ad3339640c9e0e7ca7463f641d7a2e7a7ff5dc
Status: Downloaded newer image for nginx:latest
9d77b0e0cf5d0a4a26e15c7f0216c72aa4d969c34302d217f461c7945f03d
alpine        latest    92f4e01b0c4    2 months ago  8.11MB

```

01:55:50

CLOSE SESSION

Instances

+ ADD NEW INSTANCE

192.168.0.34
model

d3iidpk6_d3iidsk69qi000ahvbagg

IP: 192.168.0.34

OPEN PORT 8080

Memory: 7.14% (285.9MB / 3.90GB)

CPU: 0.07%

SSH: ssh ip172-18-0-3-d3iidpk69qi000ahvbagg@direct.labs.play-i

DELETE EDITOR

```

$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
alpine        latest    92f4e01b0c4    2 months ago  8.11MB
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CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS   NAMES
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4d8505b0a0d0: Pull complete
0b0cf427d0d4: Pull complete
3cc5f5d3117a: Pull complete
4f40e020f03d: Pull complete
Digest: sha256:f4addd3b69e2674678ee2c7ad3339640c9e0e7ca7463f641d7a2e7a7ff5dc
Status: Downloaded newer image for nginx:latest
9d77b0e0cf5d0a4a26e15c7f0216c72aa4d969c34302d217f461c7945f03d
alpine        latest    92f4e01b0c4    2 months ago  8.11MB

```

Net secure

ip172-18-0-3-d3iidpk69qi000ahvbagg:8080.direct.labs.play-with-docker.com

School

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.

Commercial support is available at nginx.com.

Thank you for using nginx.

Task 3 – Reflection (Short Answers)

- What problem does Docker solve for developers working on Windows / macOS / Linux?

Docker abstracts away OS-level differences by packaging applications into containers that run identically across platforms. This eliminates environment mismatch issues and simplifies deployment.
- How is running an image from Docker Hub similar to installing software—but smarter?

Pulling an image from Docker Hub is like downloading a pre-configured software bundle. But it's smarter because it includes all dependencies, runs in isolation, and can be version-controlled and shared easily.

	3. How could you use Docker in future projects (AI models, APIs, web apps)? Docker can containerize: • AI models with specific Python environments and libraries (e.g., TensorFlow, PyTorch) • APIs with Flask or FastAPI, ensuring consistent behavior across dev and prod • Web apps with Node.js, React, or Django, simplifying deployment and scaling 4. Try to push a simple image to your Docker Hub account (Optional) Build a basic image: <code>docker build -t yourusername/hello-docker .</code> Push it: <pre>docker login docker push yourusername/hello-docker</pre> Deliverable Submit a 2-page PDF including: • Screenshot of your Play-with-Docker session • Screenshot of NGINX welcome page • Answers to the three reflection questions File name: B_No_StudentName_CC_A-17.pdf	
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